Combinatorics & Permutations Practice Worksheet

Questions:

1. How many 4-digit even numbers can be formed using the digits 1-9 without repetition?

2. How many different words can be formed using all the letters of the word GARDEN?

3. From a group of 7 men and 6 women, how many committees of 5 people can be formed including at least 2 women?

4. How many ways are there to distribute 5 identical balls into 3 distinct boxes?

5. A class has 14 students. In how many ways can they be divided into two groups of 7 each?

6. How many people must be present in a room so that at least two of them were born on the same day of the week?

7. How many cards must be drawn from a standard deck of 52 cards to ensure that at least three cards have the same suit?

8. How many different 3-letter words (meaningful or not) can be formed using the letters of PLANE?

9. In how many ways can 6 people be seated in a row if two specific people must not sit together?

10. How many 3-digit numbers can be formed using the digits 2, 3, 4, 5, 6, 7 without repetition?

11. How many different words can be formed using all the letters of the word BRIGHT?

12. How many 4-digit numbers can be formed using 1, 2, 3, 4, 5, 6 without repetition if the number is odd?

Answer Key:

1. 1344
2. 720
3. 1056
4. 21
5. 1716
6. 8
7. 9
8. 60
9. 480
10. 120
11. 720
12. 180